

CS502 Final Term Reference MCQ's by FAISAL DAR MIT 4th Semester

Question No: 1 (Marks: 1) - Please choose one

An optimization problem is one in which you want to find,

- ▶ Not a solution
- ▶ An algorithm
- ▶ Good solution
- ▶ **The best solution (page 97)**

Question No: 2 (Marks: 1) - Please choose one

Although it requires more complicated data structures, Prim's algorithm for a minimum spanning tree is better than Kruskal's when the graph has a large number of vertices.

- ▶ **True Not sure**
- ▶ False

Question No: 3 (Marks: 1) - Please choose one

If a problem is in NP, it must also be in P.

- ▶ **True (page 170)**
- ▶ False
- ▶ unknown

Question No: 4 (Marks: 1) - Please choose one

What is generally true of Adjacency List and Adjacency Matrix representations of graphs?

- ▶ Lists require less space than matrices but take longer to find the weight of an edge (v_1, v_2)
- ▶ **Lists require less space than matrices and they are faster to find the weight of an edge (v_1, v_2) (page 116)**

► Lists require more space than matrices and they take longer to find the weight of an edge (v_1, v_2)

► Lists require more space than matrices but are faster to find the weight of an edge (v_1, v_2)

Question No: 5 (Marks: 1) - Please choose one

If a graph has v vertices and e edges then to obtain a spanning tree we have to delete

► v edges.

► $v - e + 5$ edges

► $v + e$ edges.

► **None of these (page 125)**

Question No: 7 (Marks: 1) - Please choose one

The Huffman algorithm finds a (n) _____ solution.

► **Optimal (page 105,104)**

► Non-optimal

► Exponential

► Polynomial

Question No: 8 (Marks: 1) - Please choose one

The Huffman algorithm finds an exponential solution

► True

► **False (page 105,104)**

Question No: 9 (Marks: 1) - Please choose one

The Huffman algorithm finds a polynomial solution

► **True (page 105,104)**

► False

Question No: 10 (Marks: 1) - Please choose one

The greedy part of the Huffman encoding algorithm is to first find two nodes with larger frequency.

► True

► **False (page 100)**

Question No: 11 (Marks: 1) - Please choose one

The codeword assigned to characters by the Huffman algorithm have the property that no codeword is the postfix of any other.

► True

► **False page 101)**

Question No: 12 (Marks: 1) - Please choose one

Huffman algorithm uses a greedy approach to generate a postfix code T that minimizes the expected length $B(T)$ of the encoded string.

► True

► **False (page 101)**

Question No: 13 (Marks: 1) - Please choose one

Shortest path problems can be solved efficiently by modeling the road map as a graph.

► **True (page 153)**

► False

Question No: 14 (Marks: 1) - Please choose one

Dijkstra's single source shortest path algorithm works if all edges weights are non-negative and there are negative cost cycles.

► True

► **False (page 154)**

Question No: 15 (Marks: 1) - Please choose one

Bellman-Ford allows negative weights edges and negative cost cycles.

▶ True

▶ **False (page 154)**

Question No: 16 (Marks: 1) - Please choose one

The term “coloring” came form the original application which was in architectural design.

▶ **True not sure**

▶ False

Question No: 17 (Marks: 1) - Please choose one

In the clique cover problem, for two vertices to be in the same group, they must be adjacent to each other.

▶ **True (page 176)**

▶ False

Question No: 18 (Marks: 1) - Please choose one

Dijkstra’s algorithm is operates by maintaining a subset of vertices

▶ **True (page 155)**

▶ False

Question No: 19 (Marks: 1) - Please choose one

The difference between Prim’s algorithm and Dijkstra’s algorithm is that Dijkstra’s algorithm uses a different key.

▶ **True (page 156)**

▶ False

Question No: 21 (Marks: 1) - Please choose one

We do sorting to,

- ▶ keep elements in random positions
- ▶ keep the algorithm run in linear order
- ▶ keep the algorithm run in (log n) order
- ▶ **keep elements in increasing or decreasing order (page 36)**

Question No: 22 (Marks: 1) - Please choose one

After partitioning array in Quick sort, pivot is placed in a position such that

- ▶ **Values smaller than pivot are on left and larger than pivot are on right (page 35)**
- ▶ Values larger than pivot are on left and smaller than pivot are on right
- ▶ Pivot is the first element of array
- ▶ Pivot is the last element of array

Question No: 23 (Marks: 1) - Please choose one

Merge sort is stable sort, but not an in-place algorithm

- ▶ **True (page 54)**
- ▶ False

Question No: 24 (Marks: 1) - Please choose one

In counting sort, once we know the ranks, we simply _____ numbers to their final positions in an output array.

- ▶ Delete
- ▶ **copy (page 57)**
- ▶ Mark
- ▶ arrange

Question No: 25 (Marks: 1) - Please choose one

Dynamic programming algorithms need to store the results of intermediate sub-problems.

► **True (page 75)**

► False

Question No: 26 (Marks: 1) - Please choose one

A $p \times q$ matrix A can be multiplied with a $q \times r$ matrix B. The result will be a $p \times r$ matrix C. There are $(p \cdot r)$ total entries in C and each takes _____ to compute.

► **$O(q)$ (page 84)**

► $O(1)$

► $O(n^2)$

► $O(n^3)$

► **Using ASCII standard the string abacdaacac will be encoded with _____ bits.**

80 (page 99)

160

320

100

► **Using ASCII standard the string abacdaacac will be encoded with 160 bits.**

True

False (page 99)

► **Using ASCII standard the string abacdaacac will be encoded with 320 bits.**

True

False (page 99)

► Using ASCII standard the string abacdaacac will be encoded with 100 bits.

True

False (page 99)

The Huffman algorithm finds a (n) _____ solution.

► **Optimal (page 104,105)**

► Non-optimal

► Exponential

► Polynomial

Huffman algorithm uses a greedy approach to generate a postfix code T that minimizes the expected length B (T) of the encoded string.

► True

► **False (page 101)**

1: What is the time complexity to extract a vertex from the priority queue in Prim's algorithm?

• **$O \log (v)$ (page 152)**

• $v.v$

• $e.e$

• $\log(e)$

2: Which statement is true?

• If a dynamic-programming problem satisfies the optimal-substructure property, then a locally optimal solution is globally optimal.

• **If a greedy choice property satisfies the optimal-substructure property, then a locally optimal solution is globally optimal. (page 97)**

• both of above

- none of above

3: Kruskal's algorithm (choose best non-cycle edge) is better than Prim's (choose best tree edge) when the graph has relatively few edges.

- **True (NOT Sure)**

- False

4: You have an adjacency list for G, what is the time complexity to compute Graph transpose G^T ?

- **$V+e$ (page-116)**

- $v.e$

- v

- e

5: What general property of the list indicates that the graph has an isolated vertex?

- There is Null pointer at the end of list.
- **The Isolated vertex is not handled in list. (Not sure)**
- Only one value is entered in the list.
- There is at least one null list.

6: Which is true statement?

- Breadth first search is shortest path algorithm that works on un-weighted graphs.
- Depth first search is shortest path algorithm that works on un-weighted graphs.
- **Both of above are true. (page 153)**
- None of above are true.

7: A dense undirected graph is:

- **A graph in which $E = O(V^2)$ ref Book(page-379)**
- A graph in which $E = O(V)$

- A graph in which $E = O(\log V)$
- All items above may be used to characterize a dense undirected graph.

8: What is generally true of Adjacency List and Adjacency Matrix representations of graphs?

- Lists require less space than matrices but take longer to find the weight of an edge (v_1, v_2)
- **Lists require less space than matrices and they are faster to find the weight of an edge (v_1, v_2) Repeat**
- Lists require more space than matrices and they take longer to find the weight of an edge (v_1, v_2) .

9: Suppose that a graph $G = (V, E)$ is implemented using adjacency lists. What is the complexity of a breadth-first traversal of G ?

- $O(|V|^2)$
- **$O(|V| + |E|)$ (Not sure)**
- $O(|V|^2|E|)$
- $O(|V| + |E|)$

10: The relationship between number of back edges and number of cycles in DFS is,

- **Both are equal (rep)**
- Back edges are half of cycles
- Back edges are one quarter of cycles
- There is no relationship between no. of edges and cycles

11: Using ASCII standard the string “abacdaacacwe” will be encoded with _____

bits

- 64
- 128
- 96 rep
- 120

12: What is the time complexity to extract a vertex from the priority queue in Prim’s algorithm?

- $\log(V)$ (Page-152)
- $V.V$
- $E.E$
- $\log$

13: the analysis of selection algorithm shows the total running time is indeed-----in n .

- arithmetic
- geometric
- linear (page-37)
- orthogonal

(1) In Prim’s algorithm, the additional information maintained by the algorithm is the length of the shortest edge from vertex v to points already in the tree.

A) TRUE

B) FALSE (Not sure)

C) UNKNOWN

(2) Although it requires more complicated data structures, Prim's algorithm for a minimum spanning tree is better than Kruskal's when the graph has a large number of vertices.

A) TRUE. (Not sure)

B) FALSE

C: UNKNOWN

(3) If a problem is NP-complete, it must also be in NP.

A) TRUE. (Page-171)

B) FALSE

C) UNKNOWN

(4) Which statement is true

(I) The running time of Bellman-Ford algorithm is $T(V^2E)$

(II) Both Dijkstra's algorithm and Bellman-Ford are based on performing repeated relaxations

(III) The 0-1 knapsack problem is hard to solve

• Only I

• Only III

• Both I and III

• **All of these**

5) Which of the following arrays represent descending (max) heaps?

I. [10,7,7,2,4,6]

II. [10,7,6,2,4,7]

III. [10,6,7,2,4,6]

IV. [6,6,7,2,4,10]

• **Only II (page-140)**

- Only IV
- Both II and IV
- Both I and III

6. Which of the following statement(s) is/are correct?

- (a) $O(n \log n + n^2) = O(n^2)$.
- (b) $O(n \log n + n^2) = O(n^2 \log 2n)$
- (c) $O(c n^2) = O(n^2)$ where c is a constant.
- (d) $O(c n^2) = O(c)$ where c is a constant.
- (e) $O(c) = O(1)$ where c is a constant.

- **Only (a) & (e) (Not sure)**
- Both (c) and (e)

7. Which of the shortest path algorithms would be most appropriate for finding paths in the graph with negative edge weights and cycles?

- I. Dijkstra's Algorithm
- II. Bellman-Ford Algorithm
- III. Floyd Warshall Algorithm

- Only II
- Only III
- **Both II & III (Not sure)159**

9. Suppose we have two problems A and B .Problem A is polynomial-time reducible and problem B is NP-complete. If we reduce problem A into B then problem A becomes NP-complete

- **Yes**
- No

11. The recurrence relation of Tower of Hanoi is given below

? 1 if $n = 1$ (page 30)

$T_n = ?$

-133()

$2(T_{n-1} + 1)$ if $n > 1$

In order to move a tower of 6 rings from one peg to another, how many moves are required?

- 15
- 7
- 63
- **32**

12. Edge (u, v) is a forward edge if

- u is a proper descendant of v in the tree
- **v is a proper descendant of u in the tree (page 129)**
- None of these

13. Is $2^{2n} = O(2^n)$?

$2^n - 26$?

14. If, in a DFS forest of digraph $G = (V, E)$, $f[u] = f[v]$ for an edge $(u, v) \in E$ then the edge is called

- **Back edge (page 129)**
- Forward edge
- Cross Edge
- Tree Edge
- None of these

16. Best and worst case times of an algorithm may be same.

• **True** (not sure)

• False

17. Can an adjacency matrix for a directed graph ever not be square in shape?

• **Yes** ()

• No

1. In which order we can sort?

• increasing order only

• decreasing order only

• **increasing order or decreasing order** (sure)

• both at the same time

2. heap is a left-complete binary tree that conforms to the _____

• increasing order only

• decreasing order only

• **heap order** (page40)

• $(\log n)$ order

3. In the analysis of Selection algorithm, we make a number of passes, in fact it could be as many as,

• **$T(n)$** (page37)

• $T(n / 2)$

• $\log n$

• $n / 2 + n / 4$

4. How much time merge sort takes for an array of numbers?

• $T(n^2)$

- $T(n)$
- $T(\log n)$
- **$T(n \log n)$** (page 40)

5. One of the clever aspects of heaps is that they can be stored in arrays without using any _____.

- **pointers** (page 40)
- constants
- variables
- functions

6. the analysis of Selection algorithm, we eliminate a constant fraction of the array with each phase; we get the convergent _____ series in the analysis

- linear
- arithmetic
- **geometric** (page 37)
- exponent

7:. Sieve Technique applies to problems where we are interested in finding a single item from a larger set of _____

- **n items** (page34)
- phases
- pointers
- constant

8. The sieve technique works in _____ as follows

- **phases** (page34)
- numbers

- integers
- routines

9. For the heap sort, access to nodes involves simple _____ operations.

- **arithmetic (page 41)**

- binary
- algebraic
- logarithmic

10. The analysis of Selection algorithm shows the total running time is indeed _____ in n ,

- arithmetic
- geometric
- **linear (page 37)**
- orthogonal

11. Divide-and-conquer as breaking the problem into a small number of

- pivot
- Sieve
- **smaller sub problems (page 34)**
- Selection

12. Slow sorting algorithms run in,

- **$T(n^2)$ (page 39)**
- $T(n)$
- $T(\log n)$
- $T(n \log n)$

13. A heap is a left-complete binary tree that conforms to the

- increasing order only
- decreasing order only
- **heap order (page 40)**
- $(\log n)$ order

14. For the heap sort we store the tree nodes in

- **level-order traversal (page 40)**
- in-order traversal
- pre-order traversal
- post-order traversal

15. The reason for introducing Sieve Technique algorithm is that it illustrates a very important special case of,

- **divide-and-conquer, (page 34)**
- decrease and conquer
- greedy nature
- 2-dimension Maxima

16. We do sorting to,

Select correct option:

- keep elements in random positions
- keep the algorithm run in linear order
- keep the algorithm run in $(\log n)$ order
- **keep elements in increasing or decreasing order (sure)**

17. Sorting is one of the few problems where provable _____ bounds exists on how fast we can sort,

Select correct option:

- upper
- **lower (page 39)**
- average
- $\log n$

For the heap sort we store the tree nodes in

Select correct option:

- **level-order traversal (repeat)**
- in-order traversal
- pre-order traversal
- post-order traversal

20: In Sieve Technique we do not know which item is of interest

Select correct option:

- **True (page34)**
- False

21: Slow sorting algorithms run in,

- **$T(n^2)$ (repeat page 39)**
- $T(n)$
- $T(\log n)$
- $T(n \log n)$

22: Divide-and-conquer as breaking the problem into a small number of

- pivot

- Sieve
- **smaller sub problems (repeat)**
- Selection

23: For the sieve technique we solve the problem,

- **recursively (page 35)**
- mathematically
- precisely
- accurately

24: we do sorting to,

- keep elements in random positions
- keep the algorithm run in linear order
- keep the algorithm run in $(\log n)$ order
- **keep elements in increasing or decreasing order (repeat sure)**

25: The reason for introducing Sieve Technique algorithm is that it illustrates a very important special case of,

- **divide-and-conquer (page 34)**
- decrease and conquer
- greedy nature
- 2-dimension Maxima

26: In Sieve Technique we do

not know which item is of interest

- **true (page 34)**
- false

27: In the analysis of Selection algorithm, we make a number of passes, in fact it could be as many as,

- $T(n)$
- $T(n/2)$
- **$\log n$ (page 37)**
- $n/2 + n/4$

28: Divide-and-conquer as breaking the problem into a small number of

- pivot
- Sieve
- **smaller sub problems (repeat)**
- Selection

29: A heap is a left-complete binary tree that conforms to the _____

- increasing order only
- decreasing order only
- **heap order (repeat)**
- $(\log n)$ order

30: Slow sorting algorithms run in,

- **$T(n^2)$ (repeat)**
- $T(n)$
- $T(\log n)$
- $T(n \log n)$

31: One of the clever aspects of heaps is that they can be stored in arrays without using any _____.

- **pointers (repeat)**
- constants

- variables
- functions

32: Sorting is one of the few problems where provable _____ bounds exist on how fast we can sort,

- upper
- **lower (repeat)**
- average
- $\log n$

33: For the sieve technique we solve the problem,

- mathematically
- precisely
- accurately
- **recursively (repeat page35)**

34: Sieve Technique can be applied to selection problem?

- **True (page 35)**
- False

37: Heaps can be stored in arrays without using any pointers; this is due to the _____ nature of the binary tree,

- **left-complete (page 40)**
- right-complete
- tree nodes
- tree leaves

38: How many elements do we eliminate in each time for the Analysis of Selection algorithm?

- $n / 2$ elements
- $(n / 2) + n$ elements (page 37)
- $n / 4$ elements
- $2n$ elements

39: We do sorting to,

- keep elements in random positions
- keep the algorithm run in linear order
- keep the algorithm run in $(\log n)$ order
- keep elements in increasing or decreasing order (repeat sure)

40: In which order we can sort?

- increasing order only
- decreasing order only
- increasing order or decreasing order (repeat sure)
- both at the same time

41: : In the analysis of Selection algorithm, we make a number of passes, in fact it could be as many as,

- $T(n)$
- $T(n / 2)$
- $\log n$ (repeat page 37)
- $n / 2 + n / 4$

42: The sieve technique is a special case, where the number of sub problems is just

- 5

- Many

- **1** (pag 34)

- few

Question No: 1 (Marks: 1) - Please choose one

Random access machine or RAM is a/an

- ▶ Machine build by Al-Khwarizmi

- ▶ Mechanical machine

- ▶ Electronics machine

- ▶ **Mathematical model** (page 10)

Question No: 2 (Marks: 1) - Please choose one

_____ is a graphical representation of an algorithm

- ▶ Σ notation

- ▶ Θ notation

- ▶ **Flowchart** (sure)

- ▶ Asymptotic notation

Question No: 3 (Marks: 1) - Please choose one

A RAM is an idealized machine with _____ random-access memory.

- ▶ 256MB

- ▶ 512MB

- ▶ **an infinitely large** (page 10)

- ▶ 100GB

Question No: 4 (Marks: 1) - Please choose one

What type of instructions Random Access Machine (RAM) can execute? Choose best answer

- ▶ Algebraic and logic
- ▶ Geometric and arithmetic
- ▶ **Arithmetic and logic (page 10)**
- ▶ Parallel and recursive

Question No: 5 (Marks: 1) - Please choose one

What will be the total number of max comparisons if we run brute-force maxima algorithm with n elements?

- ▶ **n^2 (not sure)**
- ▶ $2n/n$
- ▶ n
- ▶ $8n$

Question No: 6 (Marks: 1) - Please choose one

What is the solution to the recurrence $T(n) = T(n/2) + n$.

- ▶ $O(\log n)$
- ▶ **$O(n)$ (sure)**
- ▶ $O(n \log n)$
- ▶ $O(n^2)$

Question No: 7 (Marks: 1) - Please choose one

Consider the following code:

```
For(j=1; j<n; j++)
```

```
For(k=1; k<15; k++)
```

```
For(l=5; l<n; l++)
```

```
{
```

```
Do_something_constant();  
}
```

What is the order of execution for this code.

▶ **O(n)**

▶ **O(n³) (if nested loops sure)**

▶ O(n² log n)

▶ O(n²)

Question No: 8 (Marks: 1) - Please choose one

Consider the following Algorithm:

```
Factorial (n){  
    if (n=1)  
        return 1  
    else  
        return (n * Factorial(n-1))  
}
```

Recurrence for the following algorithm is:

▶ $T(n) = T(n-1) + 1$

▶ $T(n) = nT(n-1) + 1$

▶ $T(n) = T(n-1) + n$

▶ **$T(n) = T(n(n-1)) + 1$ (not sure)**

Question No: 9 (Marks: 1) - Please choose one

What is the total time to heapify?

▶ **O(log n)** (page 43)

▶ O(n log n)

► $O(n^2 \log n)$

► $O(\log^2 n)$

Question No: 10 (Marks: 1) - Please choose one

When we call heapify then at each level the comparison performed takes time

► It will take $\Theta(1)$ (page 43)

► Time will vary according to the nature of input data

► It can not be predicted

► It will take $\Theta(\log n)$

Question No: 1 (Marks: 1) - Please choose one

An optimization problem is one in which you want to find,

► Not a solution

► An algorithm

► Good solution

► The best solution (Page 97)

Question No: 2 (Marks: 1) - Please choose one

Although it requires more complicated data structures, Prim's algorithm for a minimum spanning tree is better than Kruskal's when the graph has a large number of vertices.

► True (Not sure)

► False

Question No: 3 (Marks: 1) - Please choose one

If a problem is in NP, it must also be in P.

► True

► False (page 171)

► unknown

Question No: 4 (Marks: 1) - Please choose one

What is generally true of Adjacency List and Adjacency Matrix representations of graphs?

- ▶ Lists require less space than matrices but take longer to find the weight of an edge (v1,v2)
- ▶ **Lists require less space than matrices and they are faster to find the weight of an edge (v1,v2)**
- ▶ Lists require more space than matrices and they take longer to find the weight of an edge (v1,v2)
- ▶ Lists require more space than matrices but are faster to find the weight of an edge (v1,v2)

Question No: 5 (Marks: 1) - Please choose one

If a graph has v vertices and e edges then to obtain a spanning tree we have to delete

- ▶ v edges.
- ▶ $v - e + 5$ edges
- ▶ $v + e$ edges.
- ▶ **None of these**

Question No: 6 (Marks: 1) - Please choose one

Maximum number of vertices in a Directed Graph may be $|V|^2$

- ▶ **True**
- ▶ **False (vertices “n”)**

Question No: 7 (Marks: 1) - Please choose one

The Huffman algorithm finds a (n) _____ solution.

- ▶ **Optimal (page 105)**
- ▶ Non-optimal
- ▶ Exponential
- ▶ Polynomial

Question No: 8 (Marks: 1) - Please choose one

The Huffman algorithm finds an exponential solution

► True

► False (page 105)

Question No: 9 (Marks: 1) - Please choose one

The Huffman algorithm finds a polynomial solution

► True

► False (page 105)

Question No: 10 (Marks: 1) - Please choose one

The greedy part of the Huffman encoding algorithm is to first find two nodes with **larger** frequency.

► True

► False (page 100)

Question No: 11 (Marks: 1) - Please choose one

The codeword assigned to characters by the Huffman algorithm have the property that no codeword is the postfix of any other.

► True

► False (page 101) rep

Question No: 12 (Marks: 1) - Please choose one

Huffman algorithm uses a greedy approach to generate a postfix code T that minimizes the expected length $B(T)$ of the encoded string.

► True

► False (page 102) rep

Question No: 13 (Marks: 1) - Please choose one

Shortest path problems can be solved efficiently by modeling the road map as a graph.

► True (page 153) rep

► False

Question No: 14 (Marks: 1) - Please choose one

Dijkstra's single source shortest path algorithm works if all edges weights are non-negative and there are negative cost cycles.

► True

► **False** (page 154) rep

Question No: 15 (Marks: 1) - Please choose one

Bellman-Ford allows negative weights edges and negative cost cycles.

► True

► **False** (page 159) rep

Question No: 16 (Marks: 1) - Please choose one

The term “coloring” came from the original application which was in architectural design.

► **True** (page 179) rep

► False

Question No: 17 (Marks: 1) - Please choose one

In the clique cover problem, for two vertices to be in the same group, they must be adjacent to each other.

► **True** (page 176) rep

► False

Question No: 18 (Marks: 1) - Please choose one

Dijkstra’s algorithm is operates by maintaining a subset of vertices

► **True** (page 155) rep

► False

Question No: 19 (Marks: 1) - Please choose one

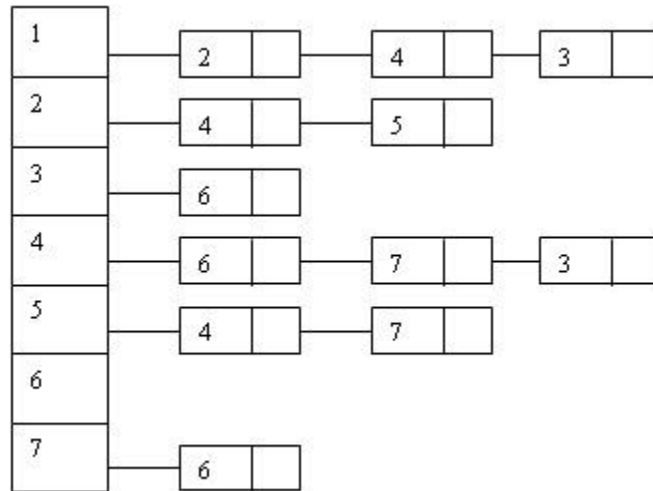
The difference between Prim’s algorithm and Dijkstra’s algorithm is that Dijkstra’s algorithm uses a different key.

► **True** (page 156) rep

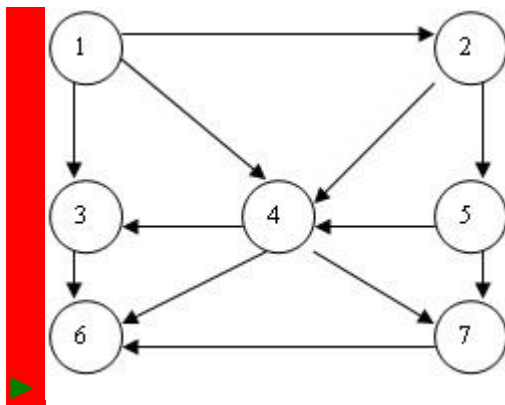
► False

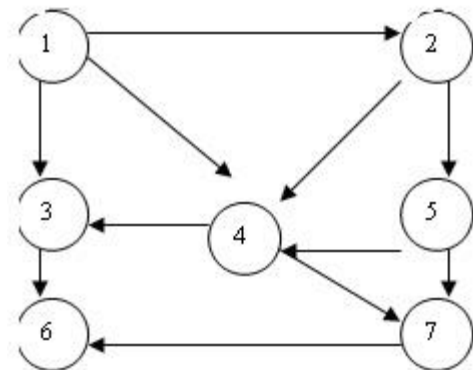
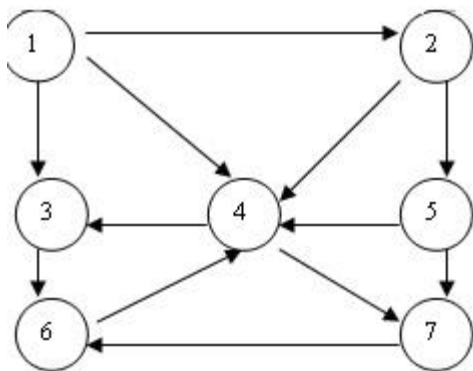
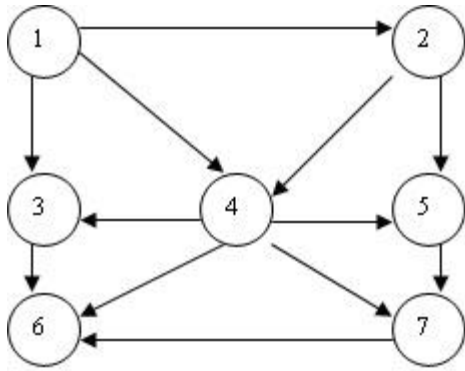
Question No: 20 (Marks: 1) - Please choose one

Consider the following adjacency list:



Which of the following graph(s) describe(s) the above adjacency list?





- ▶
- ▶
- ▶

Question No: 21 (Marks: 1) - Please choose one

We do sorting to,

- ▶ keep elements in random positions
- ▶ keep the algorithm run in linear order
- ▶ keep the algorithm run in $(\log n)$ order
- ▶ **keep elements in increasing or decreasing order**

Question No: 22 (Marks: 1) - Please choose one

After partitioning array in Quick sort, pivot is placed in a position such that

- ▶ **Values smaller than pivot are on left and larger than pivot are on right**
- ▶ Values larger than pivot are on left and smaller than pivot are on right
- ▶ **Pivot is the first element of array**
- ▶ Pivot is the last element of array

Question No: 23 (Marks: 1) - Please choose one

Merge sort is stable sort, but not an in-place algorithm

- ▶ **True** (page 54) rep
- ▶ False

Question No: 24 (Marks: 1) - Please choose one

In counting sort, once we know the ranks, we simply _____ numbers to their final positions in an output array.

- ▶ Delete
- ▶ **copy** (page 57) rep
- ▶ Mark
- ▶ arrange

Question No: 25 (Marks: 1) - Please choose one

Dynamic programming algorithms need to store the results of intermediate sub-problems.

- ▶ **True** (page 75) rep
- ▶ False

Question No: 26 (Marks: 1) - Please choose one

A $p \times q$ matrix A can be multiplied with a $q \times r$ matrix B. The result will be a $p \times r$ matrix C. There are $(p \cdot r)$ total entries in C and each takes _____ to compute.

- ▶ **O (q)** (page 84) rep
- ▶ O (1)
- ▶ O (n^2)
- ▶ O (n^3)

FINAL TERM EXAMINATION

Fall 2008

CS502- Fundamentals of Algorithms (Session - 1)

Question No: 1 (Marks: 1) - Please choose one

_____ is a graphical representation of an algorithm

- notation
- **Flowchart**
- Asymptotic notation
- notation

Question No: 2 (Marks: 1) - Please choose one

Which of the following is calculated with **big o notation**?

- Lower bounds
- **Upper bounds**
- Both upper and lower bound

- Medium bounds

Question No: 3 (Marks: 1) - Please choose one

Merge sort makes two recursive calls. Which statement is true after these recursive calls finish, but before the merge step?

- The array elements form a heap
- **Elements in each half of the array are sorted amongst themselves (page 28)**
- Elements in the first half of the array are less than or equal to elements in the second half of the array
- None of the above

Question No: 4 (Marks: 1) - Please choose one

Who invented Quick sort procedure?

- **Hoare reference book (page 95)**
- Sedgewick
- Mellroy
- Coreman

Question No: 5 (Marks: 1) - Please choose one

What is the solution to the recurrence $T(n) = T(n/2) + n$, $T(1) = 1$

- $O(\log n)$
- $O(n)$
- **$O(n \log n)$ not sure**
- $O(2n)$

Question No: 7 (Marks: 1) - Please choose one

If a graph has v vertices and e edges then to obtain a spanning tree we have to delete v edges.

- v
- $e + 5$ edges
- $v + e$ edges.
- **None of these**

Question No: 8 (Marks: 1) - Please choose one

Can an adjacency matrix for a directed graph ever not be square in shape?

- Yes
- **No**

Question No: 9 (Marks: 1) - Please choose one

One of the clever aspects of heaps is that they can be stored in arrays without using any

_____.

- **Pointers (page 40)**
- constants
- variables
- functions

Question No: 10 (Marks: 1) - Please choose one

Merge sort requires extra array storage,

- **True (page 54)**
- False

Question No: 11 (Marks: 1) - Please choose one

Non-optimal or greedy algorithm for money change takes_____

- **$O(k)$ (page 99)**

- $O(kN)$
- $O(2k)$
- $O(N)$

Question No: 12 (Marks: 1) - Please choose one

The Huffman codes provide a method of encoding data **inefficiently** when coded using ASCII standard.

- True
- **False** (page 99)

Question No: 13 (Marks: 1) - Please choose one

Using ASCII standard the string abacdaacac will be encoded with _____ bits.

- **80** (page 99)
- 160
- 320
- 100

Question No: 14 (Marks: 1) - Please choose one

Using ASCII standard the string abacdaacac will be encoded with 160 bits.

- True
- **False** (page 99)

Question No: 15 (Marks: 1) - Please choose one

Using ASCII standard the string abacdaacac will be encoded with 320 bits.

- True
- **False** (page 99)

Question No: 16 (Marks: 1) - Please choose one

Using ASCII standard the string abacdaacac will be encoded with 100 bits.

- True
- **False** (page 99)

Question No: 17 (Marks: 1) - Please choose one

Using ASCII standard the string abacdaacac will be encoded with 32 bytes

- True
- **False** (page 99)

Question No: 18 (Marks: 1) - Please choose one

The greedy part of the Huffman encoding algorithm is to first find two nodes with **smallest** frequency.

- **True** (page 100)
- False

Question No: 19 (Marks: 1) - Please choose one

The greedy part of the Huffman encoding algorithm is to first find two nodes with **character** frequency

- True
- **False** (page 100)

Question No: 20 (Marks: 1) - Please choose one

Huffman algorithm uses a greedy approach to generate an antefix code T that minimizes the expected length $B(T)$ of the encoded string.

- True
- **False** (Page 102)

Question No: 21 (Marks: 1) - Please choose one

Depth first search is shortest path algorithm that works on un-weighted graphs.

- True

False (page 153)

Question No: 22 (Marks: 1) - Please choose one

Dijkstra's single source shortest path algorithm works if all edges weights are nonnegative and there are no negative cost cycles.

- **True** (page 159)
- False

Question No: 23 (Marks: 1) - Please choose one

Dijkstra's single source shortest path algorithm works if all edges weights are negative and there are no negative cost cycles.

- True
- **False** (page 159)

Question No: 24 (Marks: 1) - Please choose one

Floyd-Warshall algorithm is a dynamic programming algorithm; the genius of the algorithm is in the clever recursive formulation of the shortest path problem.

- **True** (page 162)
- False

Question No: 25 (Marks: 1) - Please choose one

Floyd-Warshall algorithm, as in the case with DP algorithms, we avoid recursive evaluation by generating a table for

- k
- ij
- True
- False

the case with DP algorithms, we will avoid recursive evaluation by generating a table for $d(k)ij$

Question No: 26 (Marks: 1) - Please choose one

The term coloring came from the original application which was in map drawing.

- **True** (page 173)
- False

Question No: 27 (Marks: 1) - Please choose one

In the clique cover problem, for two vertices to be in the same group, they must be _____ each other.

- Apart from
- Far from
- Near to
- **Adjacent to** (Page 176)

Question No: 28 (Marks: 1) - Please choose one

In the clique cover problem, for two vertices to be in the same group, they must be apart from each other.

- True
- **False** (Page 176)

Question No: 29 (Marks: 1) - Please choose one

The difference between Prim's algorithm and Dijkstra's algorithm is that Dijkstra's algorithm uses a different key.

- **True** (Page 156)
- False

Question No: 30 (Marks: 1) - Please choose one

The difference between Prim's algorithm and Dijkstra's algorithm is that Dijkstra's algorithm uses a same key.

- True
- **False** (Page 156)

FINALTERM EXAMINATION

Spring 2010

CS502- Fundamentals of Algorithms (Session - 4)

Question No: 1 (Marks: 1) - Please choose one

An optimization problem is one in which you want to find,

Not a solution An algorithm
Good solution

The best solution Page 227 Ref book

Question No: 2 (Marks: 1) - Please choose one

Although it requires more complicated data structures, Prim's algorithm for a minimum spanning tree is better than Kruskal's when the graph has a large number of vertices.

True

False

Question No: 3 (Marks: 1) - Please choose one

If a problem is in NP, it must also be in P.

True Page 178

False

unknown

Question No: 4 (Marks: 1) - Please choose one

What is generally true of Adjacency List and Adjacency Matrix representations of graphs?

Lists require less space than matrices but take longer to find the weight of an edge (v_1, v_2)

Lists require less space than matrices and they are faster to find the weight of an edge (v_1, v_2)

Lists require more space than matrices and they take longer to find the weight of an edge (v_1, v_2)

Lists require more space than matrices but are faster to find the weight of an edge (v_1, v_2)

Question No: 5 (Marks: 1) - Please choose one

If a graph has v vertices and e edges then to obtain a spanning tree we have to delete

v edges.

$v - e + 5$ edges $v + e$ edges.

None of these

Question No: 6 (Marks: 1) - Please choose one

Maximum number of vertices in a Directed Graph may be $|V^2|$

True

False

Question No: 7 (Marks: 1) - Please choose one

The Huffman algorithm finds a (n) _____ solution.

Optimal page102

Non-optimal

Exponential

Polynomial

Question No: 8 (Marks: 1) - Please choose one

The Huffman algorithm finds an exponential solution

True

False

Question No: 9 (Marks: 1) - Please choose one

The Huffman algorithm finds a polynomial solution

True

False

Question No: 10 (Marks: 1) - Please choose one

The greedy part of the Huffman encoding algorithm is to first find two nodes with larger frequency

True

False

Question No: 11 (Marks: 1) - Please choose one

The codeword assigned to characters by the Huffman algorithm have the property that no codeword is the postfix of any other.

True

False

Page 101

Question No: 12 (Marks: 1) - Please choose one

Huffman algorithm uses a greedy approach to generate a postfix code T that minimizes the expected length $B(T)$ of the encoded string.

True

False

page 102

Question No: 13 (Marks: 1) - Please choose one

Shortest path problems can be solved efficiently by modeling the road map as a graph.

True

Page 153

False

Question No: 14 (Marks: 1) - Please choose one

Dijkstra's single source shortest path algorithm works if all edges weights are non-negative and there are negative cost cycles.

True

False

page 159

Question No: 15 (Marks: 1) - Please choose one

Bellman-Ford allows negative weights edges and negative Cost cycles.

True

False page 159

Question No: 16 (Marks: 1) - Please choose one

Question No: 16 (Marks: 1) - Please choose one

The term “coloring” came from the original application, which was in architectural design?

True

False Page 173

Which was in architectural design?

True

False Page 173

Question No: 17 (Marks: 1) - Please choose one

In the clique cover problem, for two vertices to be in the same group, they must be adjacent to each other.

True page 176

False

Question No: 18 (Marks: 1) - Please choose one

Dijkstra’s algorithm is operates by maintaining a subset of vertices

True page 155

False

Question No: 19 (Marks: 1) - Please choose one

The difference between Prim’s algorithm and Dijkstra’s algorithm is that Dijkstra’s algorithm uses a different key.

True Page 149-154

False

Question No: 21 (Marks: 1) - Please choose one

We do sorting to,

Keep elements in random positions

Keep the algorithm run in linear order keep the algorithm run in

(log n) order

Keep elements in increasing or decreasing order

Question No: 22 (Marks: 1) - Please choose one

After partitioning array in Quick sort, pivot is placed in a position such that

Values smaller than pivot are on left and larger than pivot are on right

Values larger than pivot are on left and smaller than pivot are on right

Pivot is the first element of array Pivot is the last element of array

Question No: 23 (Marks: 1) - Please choose one

Merge sort is stable sort, but not an in-place algorithm

True

False

Question No: 24 (Marks: 1) - Please choose one

In counting sort, once we know the ranks, we simply numbers to their final positions in an output array.

Delete **copy**

Page 57

Mark

arrange

Question No: 25 (Marks: 1) - Please choose one

Dynamic programming algorithms need to store the results of intermediate sub-problems.

True Page 75

False

CS502 - Fundamentals of Algorithms

Quiz No.3 Dated 28-01-2013

**In in-place sorting algorithm is one that uses arrays for storage:
An additional array**

No additional array (page-54)

Both of above may be true according to algorithm

More than 3 arrays of one dimension.

The running time of quick sort depends heavily on the selection of

No of inputs

Arrangement of elements in array

Size o elements

Pivot element (page-49)

In stable sorting algorithm

One array is used

In which duplicating elements are not handled.

More then one arrays are required.

Duplicating elements remain in same relative position after sorting. (Page-54)

Which sorting algorithm is faster:

$O(n^2)$

$O(n \log n)$

O(n+k) (Right Answer) Not sure

O(n³)

In Quick sort algorithm, constants hidden in T(n lg n) are

Large

Medium

Not known

Small (ref.Book.page-95)

Quick sort is based on divide and conquer paradigm; we divide the problem on base of pivot element and:

There is explicit combine process as well to conquer the solution. (Page-96)

No work is needed to combine the sub-arrays, the array is already sorted

Merging the sub arrays

None of above.

There is relationship between number of back edges and number of cycles in DFS

Select correct option:

Both are equal. (Page-131)

Cycles are half of back edges.

Cycles are one fourth of back edges.

There is no relationship between back edges and number of cycle

You have an adjacency list for G, what is the time complexity to compute Graph Transpose G^T ?

(V+E) (Page-138)

V.E

V

E

What is the time complexity to extract a vertex from the priority queue in Prim's Algorithm?

log (V) (page-152)

V.V

E.E

log (E)

Dijkstra's algorithm:

Has greedy approach to find all shortest paths

Has both greedy and dynamic approach to find all shortest paths

Has greedy approach to compute single source shortest paths to all other vertices (page-154)

Has both greedy and dynamic approach to compute single source shortest paths to all other vertices.

What algorithm technique is used in the implementation of Kruskal solution for the MST?

Greedy Technique (Page-142)

Divide-and-Conquer Technique

Dynamic Programming Technique

The algorithm combines more than one of the above techniques

What is the time complexity to extract a vertex from the priority queue in Prim's algorithm?

$O(\log E)$

? (V)

? (V+E)

$O(\log V)$ (Page-152)

Which is true statement in the following?

Kruskal algorithm is multiple source technique for finding MST.

Kruskal's algorithm is used to find minimum spanning tree of a graph, time complexity of this algorithm is $O(EV)$

Both of above

Kruskal's algorithm (choose best non-cycle edge) is better than Prim's (choose best Tree edge) when the graph has relatively few edges) (page-149)Not sure

The relationship between number of back edges and number of cycles in DFS is,

Both are equal (rep)

Back edges are half of cycles

Back edges are one quarter of cycles

There is no relationship between no. of edges and cycles (Right Answer)

Kruskal's algorithm (choose best non-cycle edge) is better than Prim's (choose best tree Edge) when the graph has relatively few edges.

True (page-149)

False

What is the time complexity to extract a vertex from the priority queue in Prim's Algorithm?

$\log(V)$ (page-152)

$V.V$

$E.E$

$\log(E)$

Suppose that a graph $G = (V, E)$ is implemented using adjacency lists. What is the complexity of a breadth-first traversal of G ?

$O(|V|^2)$

$O(|V| + |E|)$ (Right Answer) Not Sure

$O(|V|^2|E|)$

$O(|V| + |E|)$

What is generally true of Adjacency List and Adjacency Matrix representations of graphs?

Select correct option:

Lists require less space than matrices but take longer to find the weight of an edge (v_1, v_2)

Lists require less space than matrices and they are faster to find the weight of an edge (v_1, v_2) (Right Answer)

Lists require more space than matrices and they take longer to find the weight of an edge (v_1, v_2)

Lists require more space than matrices but are faster to find the weight of an edge (v_1, v_2)

What general property of the list indicates that the graph has an isolated vertex?

There is Null pointer at the end of list.

The Isolated vertex is not handled in list. (Right Answer)(Not Sure)

Only one value is entered in the list.

There is at least one null list.

A dense undirected graph is:

A graph in which $E = O(V^2)$ (Ref Book.page-379)

A graph in which $E = O(V)$

A graph in which $E = O(\log V)$

All items above may be used to characterize a dense undirected graph

In digraph $G=(V,E)$;G has cycle if and only if

The DFS forest has forward edge.

The DFS forest has back edge (page-130)

The DFS forest has both back and forward edge

BFS forest has forward edge

Back edge is:

(u, v) where v is an ancestor of u in the tree. (page-128)

(u,v) where u is an ancestor of v in the tree.

(u, v) where v is an predecessor of u in the tree.

None of above

Using ASCII standard the string “abacdaacacwe” will be encoded with _____ bits

64

128

96(Right Answer) 12characters*8bit=96

120

Cross edge is:

(u, v) where u and v are not ancestor of one another

(u, v) where u is ancestor of v and v is not descendent of u.

(u, v) where u and v are not ancestor or descendent of one another (page-129)

(u, v) where u and v are either ancestor or descendent of one another.

Which statement is true?

If a dynamic-programming problem satisfies the optimal-substructure property, then a locally optimal solution is globally optimal.

If a greedy choice property satisfies the optimal-substructure property, then a locally optimal solution is globally optimal.

Both of above (Right Answer) Not sure

None of above

**If you find yourself in maze the better traversal approach will be
A dense undirected graph is:**

A graph in which $E = O(V^2)$ (Rep)

A graph in which $E = O(V)$

A graph in which $E = O(\log V)$

All items above may be used to characterize a dense undirected graph

Which is true statement?

Breadth first search is shortest path algorithm that works on un-weighted graphs (Page-153)

Depth first search is shortest path algorithm that works on un-weighted graphs.

Both of above are true.

None of above are true.

Forward edge is:

(u, v) where u is a proper descendent of v in the tree.

(u, v) where v is a proper descendent of u in the tree. (Page-129)

(u, v) where v is a proper ancestor of u in the tree.

(u, v) where u is a proper ancestor of v in the tree.

Back edge is:

(u, v) where v is an ancestor of u in the tree. (Page-128)

(u, v) where u is an ancestor of v in the tree.

(u, v) where v is an predecessor of u in the tree.

None of above

Suppose that a graph $G = (V, E)$ is implemented using adjacency lists. What is the complexity of a breadth-first traversal of G?

$O(|V|^2)$

$O(|V| + |E|)$ (Right Answer) Not Sure

$O(|V|^2|E|)$

$O(|V| + |E|)$

In digraph $G=(V,E)$;G has cycle if and only if

The DFS forest has forward edge.

The DFS forest has back edge (Page-130)

The DFS forest has both back and forward edge

BFS forest has forward edge

What general property of the list indicates that the graph has an isolated vertex?

There is Null pointer at the end of list.

The Isolated vertex is not handled in list. (Not Sure)

Only one value is entered in the list.

There is at least one null list.

If you find yourself in maze the better traversal approach will be:

BFS

BFS and DFS both are valid (page-119)

Level order

DFS

Cross edge is :

(u, v) where u and v are not ancestor of one another

(u, v) where u is ancestor of v and v is not descendent of u.

(u, v) where u and v are not ancestor or descendent of one another (Page-129)

(u, v) where u and v are either ancestor or descendent of one another.

What algorithm technique is used in the implementation of Kruskal solution for the MST?

Greedy Technique rep (page-142)

Divide-and-Conquer Technique

Dynamic Programming Technique

The algorithm combines more than one of the above techniques

Kruskal's algorithm (choose best non-cycle edge) is better than Prim's (choose best tree edge) when the graph has relatively few

True (page-149)

False

You have an adjacency list for G , what is the time complexity to compute Graph transpose G^T ?

?($V + E$) ref.Book(Page-323)

? ($V E$)

? (V)

? (V^2)

A digraph is strongly connected under what condition?

A digraph is strongly connected if for every pair of vertices $u, v \in V$, u can reach v .

A digraph is strongly connected if for every pair of vertices $u, v \in V$, u can reach v and vice versa. (page-135)

A digraph is strongly connected if for at least one pair of vertex $u, v \in V$, u can reach v and vice versa.

A digraph is strongly connected if at least one third pair of vertices $u, v \in V$, u can reach v and vice versa.

The relationship between number of back edges and number of cycles in DFS is,

Both are equal (page-131)

Back edges are half of cycles

Back edges are one quarter of cycles

There is no relationship between no. of edges and cycles

“There is some simple relationship between the number of back edges and the number of cycles.”

What algorithm technique is used in the implementation of Kruskal solution for the MST?

Greedy Technique rep (page-142)

Divide-and-Conquer Technique

Dynamic Programming Technique

The algorithm combines more than one of the above techniques



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Reference MCQ's For Final TERM EXAMS

CS502- Design and Analysis of Algorithms